

Chapter – 1: Locating Places on the Earth?

The globe of the Earth stands in space, made up of water, earth, fire, and air is spherical... It is surrounded by all creatures, terrestrial as well as aquatic. – Aryabhata (about 500 CE)

- Imagine – visiting a city – 1st time – how to find a place?
- Ask a local – OR – look at the map
- Previous classes – learnt little about maps – this chapter – study in more detail
- A game –
 - Examine the map – small city
 - Imagine – get off from the train – railway station – go to bank – how to go? – are there diff. ways?
 - Locate – garden, school, museum
 - Bank to market – which way to go
 - Here – map comes in handy



- Map – treasure guide – where things are – how to get them
- 4 arrows – on top right corner – directions

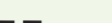
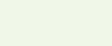
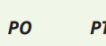
A Map and Its Components

- Map – representation / drawing – some area –
 - Small area – village, town, etc.
 - Bigger area – district or state
 - Very large area – India or even the whole world
- Map – viewed from top
- Atlas – collection of maps
- Maps – several kinds –
 - **Physical** – mainly – natural features – mountains, oceans, rivers
 - **Political** – mainly – details – countries, states, boundaries, cities, etc.
 - **Thematic** – specific kind of information
- 3 imp. components of maps – distance, direction, symbols
- Huge place – fit into small paper – how? – all thanks to **scale**
- Each cm of the printed map – represents certain distance on the ground – say – 500 m – scale: 1 cm = 500 m
- Map of India – chapter 5 – scale – bottom left corner – ‘500’ above the length – ‘km’ on the side
- This means – ruler – 2.5 cm on the map – 500 km on the ground
- Actual distance – b/w 2 points – on map – depends on scale
- Four arrows – top right – small city map
- 4 **directions** – north (top) – clockwise – east, south, west – **cardinal directions**
- Intermediate directions – northeast (NE), southeast (SE), northwest (NW), southwest (SW)



- Most maps – 1 arrow – marked with ‘N’ at the top

- **Symbols** – our map – small drawings – actual buildings – few other elements
- BUT – not enough space – map – large city or country
- Instead – symbols are used – represent these features –
 - Diff. kind of buildings – railway station, school, post office
 - Railway lines, roads
 - Natural elements – river, pond, forest

Railway Line: broad gauge, metre gauge, railway station			
Roads: metalled, unmetalled			
Boundary: international, state, district			
River, well, tank, canal, bridge			
Temple, church, mosque, chhatra			
Post Office, Post & Telegraph Office, Police Station			
Settlement, graveyard			
Trees, grass			

- This way – lots of details – limited space
- Maps – easier to read – specific symbols
- Diff. countries – diff. symbols
- Survey of India – govt. body – fixed – set of symbols

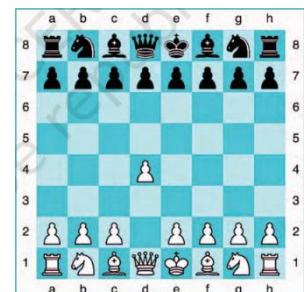
Mapping the Earth

- Mapping the Earth – little more difficult – our planet – not a flat surface
- Shape – nearly ‘sphere’ – not perfect – slightly flattened – at the poles
- Representing sphere – on a paper – not possible
- Try –
 - Peel an orange – 3 or 4 large pieces – try and flatten them
 - Cannot do it – without tearing at the edges
- **Globe** – sphere – map drawn on it – any map – the Earth, the Moon, the Mars, the stars and constellations, etc.
- Physical obj. – in the img. – sphere – made of metal, plastic or cardboard
- This chapter – study globe – same shape as Earth
- Globe – represent better geography of Earth



Understanding Coordinates

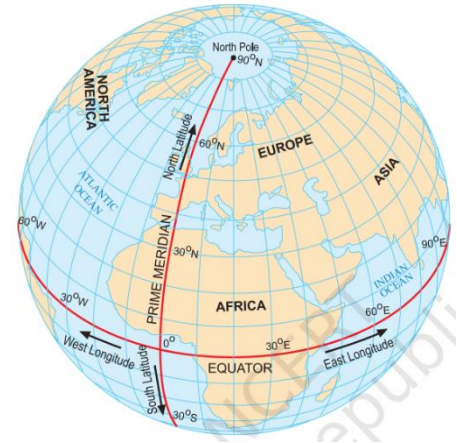
- Imagine – big market – city / town – neat rows of shops – same size
- Meet a friend – stationary shop – friend – no idea – where the shop is?
- Give directions – 7th shop in 5th row – help friend – find the shop
- Same with chess – adv. players – record moves
- Letters used – a to h – along main pieces – numbers used – 1 to 8 – on the sides
- This system – help players – mark each square – record every move
- White side – opened the game – queen’s pawn – 2 steps forward – d2 to d4 – most common move
- This system – system of **coordinates** – 2 coordinates – help to locate – shop in market – AND – square on the chessboard
- Similar system – maps use – locate place on the map



Latitudes

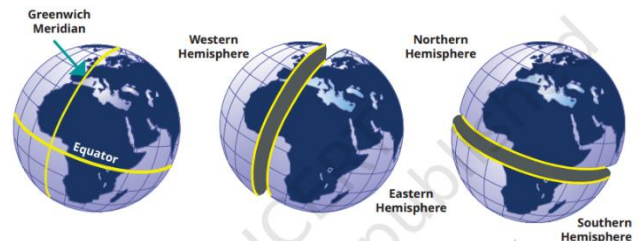
- Globe – easy to identify – north and south pole

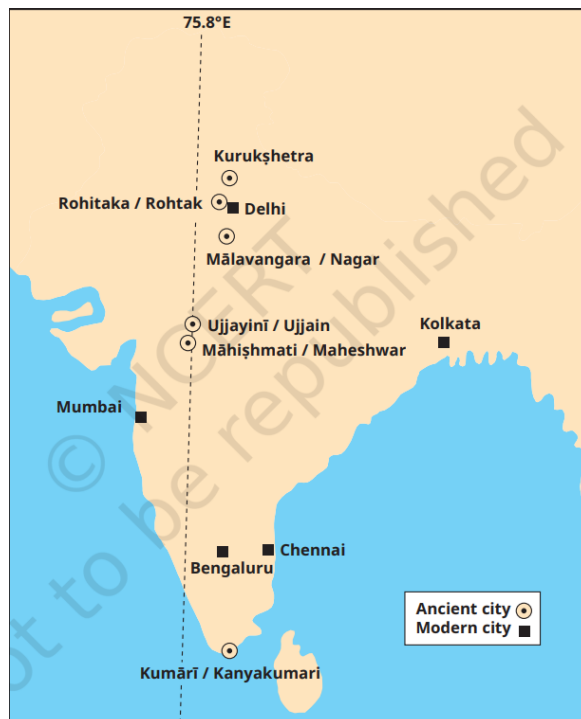
- Rotate the globe – fixed points at top and bottom – 2 poles – halfway – equator
- Imagine – standing on the equator – walk towards the poles – dist. from equator increases
- This dist. – **latitude**
- Any point – draw imaginary line – east to west – parallel to equator – **parallel of latitude** – completes a circle around the Earth
- Largest circle – equator – move northward OR southward – parallels of latitude – decrease in size
- Latitudes – expressed in **degrees** –
 - Equator – 0°
 - North pole – 90° North (90° N)
 - South pole – 90° South (90° S)
- Connection b/w latitude and climate –
 - Around the equator – climate is hot (torrid)
 - Walk towards the poles – latitude increases – climate gets moderate (temperate)
 - Closer to the poles – climate gets colder (rigid)



Longitudes

- Imagine – travel from the poles to the equator – shortest path
- Through Europe and Africa – BUT – another way – through Asia – same dist.
- These paths – **meridians of longitude** – half circles – from one pole to the other
- Earth – spins on its axis
- Place a desk lamp – bit away from the globe – focused on it – acts like the sun
- Rotate the globe slowly – eastward – at any pt. – some places – sun shines – other places – doesn't shine
- Measuring the longitude – helps in measuring the time
- Measure longitudes – define a reference pt. – **Prime Meridian**
- Also called – Greenwich Meridian – 1884 – some nations – decided – meridian through Greenwich, London – international std. – Prime Meridian – marked – 0° longitude
- Latitude – measure – dist. from equator to pole
- **Longitude** – measure – dist. from prime meridian – travel along the equator – measured in degrees
- Westward to east ward – increases from 0° to 180° – letters W or E added
- Example –
 - New York – 74° W
 - Delhi – 77° E
 - Tokyo – 140° E
- 180° W and 180° E – same line – noted as 180°
- Latitude and longitude – together – 2 **coordinates** – locate any place
- Example – Delhi – 29° N latitude and 77° E longitude
- All these lines – together – create a grid – called as grid lines
- Ancient times – India – own prime meridian – *Madhya Rekha* – passing through Ujjayini (Ujjain)
- Indian astronomers – were aware – concept – latitude, longitude and prime meridian – Ujjayini prime meridian – used in many calculations in ancient Indian texts
- India – spans from 8° N to 37° N – AND – 68° E to 97° E





Understanding Time Zones

- Spin the globe – west to east – similar to Earth – full turn – 24 hours
- Full turn – 360° – 15° per hour
- Mark the meridians every 15° – moving eastward – 0° , 15° E, 30° E, 45° E and so on
- This – same as – adding 1 hr of **local time** – 12 pm at Greenwich – 1 pm local time at 15° E, 2 pm local time at 30° E and so on
- Westward – other way round – 11 am local time at 15° W, 10 am local time at 30° W and so on
- Same method – used to calculate local time – any place
- Not convenient – any country – diff. local time
- Most countries – adopt – std. time – India Standard Time (IST) – 5 hrs 30 min ahead of Local time at Greenwich (**Greenwich Meridian Time** or GMT)
- Lines dividing time zones – not straight – respect each country's std. time – follow international borders
- Numbers inside countries – add to GMT for local time if its +ve – subtract from GMT for local time if its -ve
- Prime meridian – fixed at Greenwich – opposite line – 180° longitude – **International Date Line**
- +12 and -12 touch at 180° – crossing it – change date in the watch
- Travel eastward – subtract a day – Monday to Sunday
- Travel westward – add a day – Monday to Tuesday

